

Dual Power Automatic Transfer Switch (ATS) User Manual

Model # AB21

This product features a dual-input, single-output design that automatically selects the higher-voltage input to power the load. An integrated diode isolation protection circuit effectively prevents reverse current between the two input sources, providing electrical isolation and ensuring that the two power sources do not interfere with each other. A digital display shows the real-time output voltage.

Specification

Model	Voltage	Max Current	Transfer time	Display
AB21A	5-30V	50A	Seamless	Yes
AB21B	5-60V			

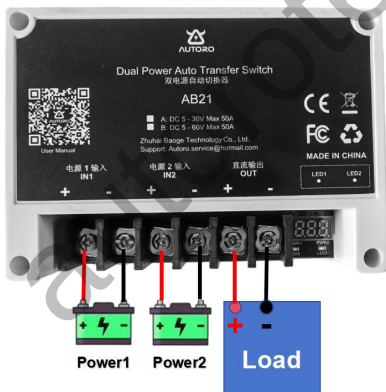
Indicator Light Descriptions

- **Red LED on:** Power source 1 is supplying the load.
- **Blue LED on:** Power source 2 is supplying the load.
- **Red and Blue LEDs both on:** Both power sources are supplying the load together (parallel mode, when input voltages are equal).

Automatic Switching Logic

- **When the two inputs have a voltage difference:** the module automatically selects the higher-voltage source to power the load. The switching process is seamless and transparent to the load, and the corresponding indicator light turns on.
- **When the two inputs have equal voltage (commonly used in dual-battery parallel expansion scenarios):** both sources output simultaneously, sharing the load, and both the Red and Blue LEDs turn on at the same time.

Wiring Diagram



PWR1: Power1 Output Indicator
PWR2: Power2 Output Indicator
Digital Tube: Voltage Display
Max Current: 50A

Contact & Support

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双电源自动切换器用户手册

型号：AB21

该产品采用双电源输入与单电源输出设计，可自动选择电压较高的一路为负载供电。内置二极管隔离保护电路，有效防止两路输入电源之间的电流倒灌，实现电气隔离，确保两路电源互不干扰。数码管实时显示当前输出电压值。

规格参数

型号	电压	最大电流	切换时间	显示功能
AB21A	5-30V	50A	0 延时	支持
AB21B	5-60V			

指示灯说明

- **红灯亮：**电源 1 单独供电。
- **蓝灯亮：**电源 2 单独供电。
- **红灯和蓝灯同时亮：**两路电源共同供电（电压相等时的并联状态）。

自动切换逻辑

- 两路输入存在压差时：自动选择电压较高的一路单独输出，切换过程无缝衔接，负载无感知，同时对应指示灯亮起。
- 两路输入电压相等时（常见于双电池并联扩容场景）：两路同时输出，共同向负载供电，红灯和蓝灯同时亮起。

接线图



PWR1：电源1输出指示灯
PWR2：电源2输出指示灯
数码管：电压显示
最大电流：50A

联系与支持

制造商：珠海宝格科技有限公司

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